

Electric Vehicle Performance and Charging Analysis

1. PROJECT DESCRIPTION

System Overview

This project analyzes electric vehicle charging behavior, battery performance, and energy consumption using data visualization techniques. The system uses interactive dashboards created in Tableau to transform EV data into meaningful insights for decision-making.

Problem Domain

With the increasing adoption of electric vehicles, it has become important to understand charging demand, battery degradation patterns, and energy consumption across cities. Without proper analysis, it becomes difficult for planners and users to optimize EV usage.

Core Technology Used

- Tableau Desktop Public Edition
- Data Visualization Techniques
- Data Analytics

Key Capabilities

- Visualizing EV charging patterns
- Monitoring battery degradation
- Comparing efficiency of EV models
- Analyzing energy consumption across cities

Target Users

- City planners

- EV fleet managers
- EV consumers
- Energy analysts

Real-world Relevance

The insights generated from this system help improve EV infrastructure planning, battery management strategies, and energy optimization.

2. APPLICATION SCENARIOS / USE CASES

Scenario 1 – Urban Energy Planning

City planners can analyze charging demand and peak charging hours to ensure grid stability.

Scenario 2 – EV Fleet Management

Fleet managers can monitor battery health and detect vehicles that require maintenance.

Scenario 3 – Consumer Decision Making

Consumers can compare EV models based on efficiency and energy usage before purchasing.

Scenario 4 – Environmental Impact Analysis

Researchers can evaluate energy consumption and sustainability benefits of EV adoption.

3. TECHNICAL ARCHITECTURE OVERVIEW

High-level Architecture

Dataset → Data Cleaning → Tableau Visualization → Interactive Dashboard → Insights

Component Interaction

- 1 Data Collection
- 2 Data Cleaning
- 3 Data Visualization
- 4 Dashboard Integration
- 5 Insight Generation

Data Flow

EV Dataset → Tableau Processing → Charts & Visualizations → Dashboard → User Insights

System Layers

Frontend Layer

Tableau dashboard interface where users interact with charts and filters.

Backend Layer

Data processing and visualization engine within Tableau.

Database Layer

EV dataset stored in CSV format.

Deployment Environment

Local system running Tableau Desktop.

4. PREREQUISITES

4.1 Software Requirements

- Tableau Desktop Public Edition
- Microsoft Excel / CSV Dataset
- Operating System (Windows / macOS)

4.2 Libraries / Frameworks / Dependencies

- Tableau Visualization Engine

4.3 Hardware Requirements

- Laptop or desktop computer
- Minimum 4GB RAM
- Internet connection (for sharing dashboards)

5. PRIOR KNOWLEDGE REQUIRED

- Basic data analytics knowledge
- Data visualization concepts
- Tableau fundamentals
- Understanding of electric vehicle concepts

6. PROJECT OBJECTIVES

Technical Objectives

- Build interactive Tableau dashboards for EV analysis.

Performance Objectives

- Efficiently analyze large EV datasets.

Deployment Objectives

- Present dashboards for visualization and decision-making.

Learning Outcomes

- Understanding data visualization techniques.
- Learning how to analyze EV performance data.

7. SYSTEM WORKFLOW

1. User loads EV dataset into Tableau.

2. Dataset is cleaned and validated.
3. Charts and visualizations are created.
4. Multiple sheets are combined into dashboards.
5. Insights are generated from visual analysis.
6. Users interact with the dashboard to explore trends.

8. MILESTONE 1: REQUIREMENT ANALYSIS & SYSTEM DESIGN

8.1 Problem Definition

Understanding EV charging patterns and battery performance using data analytics.

8.2 Functional Requirements

- Load EV dataset
- Generate visualizations
- Create dashboards
- Provide insights from charts

8.3 Non-Functional Requirements

- Fast dashboard performance
- Clear visualizations
- Easy user interaction

8.4 System Design Decisions

Tableau was selected because it allows easy creation of interactive dashboards and powerful data visualization.

8.5 Technology Stack

- Tableau Desktop
- CSV Dataset
- Data Visualization Techniques

9. MILESTONE 2: ENVIRONMENT SETUP

9.1 Development Environment Setup

Install Tableau Desktop Public Edition.

9.2 Dependency Installation

Import EV dataset into Tableau.

9.3 Project Structure Creation

Workbook contains:

- Battery Health Analysis Sheet
- Charging Pattern Sheet
- Energy Analysis Sheet
- Efficiency Analysis Sheet
- Dashboard

9.4 Configuration Setup

Data fields configured for dimensions and measures in Tableau.

10. MILESTONE 3: CORE SYSTEM DEVELOPMENT

Feature 1 – Battery Health Analysis

Scatter plots were used to analyze the relationship between battery temperature and degradation rate.

Feature 2 – Charging Pattern Analysis

Heatmap visualization shows peak charging hours across cities.

Feature 3 – Efficiency Comparison

Scatter plot compares EV model efficiency and degradation rate.

Additional Visualizations:

- Energy consumption bar chart
- Charging mode distribution pie chart
- Battery health trend line chart

11. MILESTONE 4: INTEGRATION & OPTIMIZATION

Component Integration

All sheets were integrated into a single interactive dashboard.

Performance Optimization

Efficient data aggregation techniques were used.

Security Enhancements

No sensitive data used in the dataset.

Error Handling

Data validation performed before visualization.

12. MILESTONE 5: TESTING & VALIDATION

13.1 Test Cases

Test Case	Input	Expected Output	Status
Load dataset	EV dataset	Data visible in Tableau	Pass
Generate charts	Visualization fields	Charts generated	Pass
Dashboard view	Combined sheets	Interactive dashboard	Pass

12.2 Unit Testing

Individual charts were tested separately.

12.3 Integration Testing

Multiple charts combined successfully in dashboard.

12.4 User Acceptance Testing

Dashboard reviewed by users for clarity.

12.5 Performance Metrics

Dashboard loads quickly and visualizations update dynamically.

13. DEPLOYMENT

Deployment Architecture

Tableau dashboard deployed locally and shared via Tableau Public.

Hosting Platform

Tableau Public

Deployment Steps

- 1 Import dataset
- 2 Create visualizations
- 3 Combine sheets into dashboard
- 4 Publish dashboard online

Production Considerations

Dashboard can be updated with new EV datasets for continuous analysis.

14. PROJECT STRUCTURE

EV_Project/

Dataset/
EV_Data.csv

Tableau_Workbook/
EV_Analysis.twbx

Documentation/
Project_Report.docx

Media/
Dashboard_Screenshots

15. RESULTS

System Output

Interactive Tableau dashboard displaying EV charging and battery performance data.

Performance Evaluation

Dashboard successfully visualizes patterns in EV usage and battery behavior.

Screenshots

Include dashboard images.

16. ADVANTAGES & LIMITATIONS

Advantages

- Interactive data visualization
- Easy interpretation of complex data
- Supports data-driven decisions

Limitations

- Limited to available dataset
- Real-time data integration not included

17. FUTURE ENHANCEMENTS

- Integration with real-time EV charging data
- Machine learning predictions for battery life
- Mobile dashboard access
- Cloud deployment

18. CONCLUSION

This project demonstrates how Tableau dashboards can be used to analyze electric vehicle charging behavior and battery performance. The insights generated can support

infrastructure planning, improve EV efficiency, and help consumers make informed decisions.

19. APPENDIX

- [Tableau Workbook \(.twbx\)](#)
- [EV Dataset](#)
- [Project Video Link](#)
- [GitHub Repository](#)
- [Live Demo](#)